Problem 15.9

In the previous problem, if the PhD student saves the income from her grandmother in an account also paying 3.5% effective annual interest, how much will she have when she receives the final $1,000 payment?

Solution

The student receives $1,000 each year for ten years and saves each payment in an account earning an annual effective interest rate of

$$i = 0.035.$$

We want the accumulated value at the time she receives the final payment. Thus, we use the accumulated value of an annuity-immediate:

$$Y = 1000 s_{\overline{10}|}.$$

Using the formula

$$s_{\overline{10}|} = \frac{(1+i)^{10} - 1}{i},$$

we get

$$Y = 1000 \left(\frac{(1.035)^{10} - 1}{0.035} \right).$$

Finally,

$$Y \approx 11731.39316.$$

Hence, when she receives the final $1,000 payment, she will have approximately

$$\boxed{\$11{,}731.39}.$$